

Alessio Salvatore

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Education

ETH Zürich, Zürich, Switzerland Sep 2025–Present
M.Sc. in Robotics

Coursework: Probabilistic Artificial Intelligence, Advanced Machine Learning, Image Analysis and Computer Vision, Planning and Decision Making for Autonomous Robots, 3D Vision, Perception and Learning for Robotics, Deep Generative Models, Robot Learning

Politecnico di Milano, Milan, Italy Sep 2022–Jul 2025
B.Sc. in Automation Engineering

Final Grade: 110/110 *cum laude* (equivalent GPA $\approx$ 4.0/4.0; 29.57/30).

Illinois Institute of Technology, Chicago, IL, USA Aug–Dec 2024
Exchange — Global E3 Program

Projects

Research Projects

Goal-Driven Adaptive-Resolution 3D Mapping with RL Feb–Jun 2026
Computer Vision and Geometry Group – Marc Pollefeys Lab

- Trained a PPO policy that adapts per-region voxel resolution (split/keep/merge) online during navigation under a hard memory budget, reformulating an intractable per-block action space into a compact two-threshold policy
- Re-implemented PEANUT's dense voxel map as a sparse hierarchical multi-resolution grid (Habitat, HM3D); the policy retains 79% of point-goal success with 87% less memory than a uniformly fine map (20 unseen episodes)

Knowledge Distillation for Aerial Robots Control  Video Feb–Jun 2026
Autonomous Systems Lab – Roland Siegwart Lab

- Distilled model-based controllers for an overactuated tilt-rotor MAV (OMAV) into a single recurrent LSTM policy, trained closed-loop through differentiable dynamics with DAGger on ~ 42 h of simulated flight
- In simulation the policy outperforms the nominal model-based controller in velocity tracking ($p < 10^{-10}$, 2,000 paired rollouts) and nearly halves divergence under $\pm 30\%$ model mismatch; deployed zero-shot on the real OMAV

Selected Projects

Autonomous Exploration on a Unitree A2 Quadruped  Repo 06/2026
ETH Robotics Summer School (ETHZ-RobotX)

- Deployed an autonomous search-and-explore mission with our team on a real Unitree A2 quadruped across three field sites; developed the artifact-mapping node fusing YOLO detections into a deduplicated object log

Vision-Language-Action Policies on a Real SO-101 Arm  Repo 05/2026
Project of the Class Robot Learning

- Fine-tuned SmoVLA policies for language-conditioned pick-and-place across three tasks with on-robot inference at 10 FPS; built the LeRobot data pipeline and automated H100 cloud training

Person Following on the Agibot X2 Humanoid  Repo 05/2026
RoboHack Hackathon

- Built real-time person following running fully onboard in a hackathon weekend: YOLOv8 with stereo disparity depth, Ruckig head tracking, and a state-machine locomotion supervisor holding a set distance

Work Experience

JOINT Lab — Italian Institute of Technology, Bergamo Feb–Jun 2025
Research Intern — Industrial Robotics  Repo

- Designed a linear actuator in CAD, implemented real-time control on Arduino, and validated system dynamics in MATLAB

ETH Robotics Club Nov 2025–Present
Partnerships Team Lead

- Secured sponsorships for club initiatives and events

Laelaps AI Jun 2026–Present
Robotics Engineer

- Developing multi-agent systems for robot orchestration in complex real-world environments